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Engg. Math. H.W - 2.

1. $r(x) = r_0 e^{-\lambda x}$. [$x \longrightarrow$ depth of penetration]

Fickian diffusion is the governing law for this pseudo steady state condition:

$D \cdot \frac{d^2}{dx^2}[r] = -r(x) = -r_0 e^{-\lambda x}$.

$\left[\frac{D^2}{Dx^2}\left(e^{-\lambda x}\right) = (-\lambda)^2 \cdot e^{-\lambda x} \right]$

$D \cdot \partial^2 [r] + \frac{r_0}{\lambda^2} \partial^2 (e^{-\lambda x}) = 0$.

$\partial^2 [r] + \left(\frac{r_0}{\lambda^2 D}\right) \cdot \partial^2 (e^{-\lambda x}) = 0$.

$\cancel{\left[\partial[r] + i\sqrt{\frac{r_0}{\lambda^2 D}} \cdot \partial(e^{-\lambda x})\right]\left[\partial[r] - i\sqrt{\frac{r_0}{\lambda^2 D}} \cdot \partial(e^{-\lambda x})\right] = 0}$.

$\Rightarrow \partial^2 \left[[r] + \frac{r_0}{\lambda^2 D} \cdot e^{-\lambda x} \right] = 0$.

$\therefore [r] + \frac{r_0}{\lambda^2 D} e^{-\lambda x} = Ax + B$. ...... (i).

Now use dirichlet & vonNeumann boundary conditions i.e $\boxed{C(0) = 0}$ & $\frac{\partial}{\partial x}[r] \Big/ \frac{\partial}{\partial x} C \Big|_{x=d} = 0$.

$\Rightarrow 0 + \frac{r_0}{\lambda^2 D} = B$. ...... (ii)

$\frac{\partial}{\partial x}[r] + \frac{r_0}{\lambda^2 D} \cdot (-\lambda) e^{-\lambda x} = A$.

$\Rightarrow 0 + \frac{r_0}{\lambda^2 D} \cdot (-\lambda) e^{-\lambda d} = A$ ...... (iii)

$\therefore A = -\frac{r_0}{\lambda D} e^{-\lambda d}$, $B = \frac{r_0}{\lambda^2 D}$.

Combining (i), (ii) & (iii) :

$$\therefore [r] + \frac{r_0}{\lambda^2 D} e^{-\lambda x} = -\frac{r_0}{\lambda D} e^{-\lambda d} x + \frac{r_0}{\lambda^2 D}$$

$$\therefore [r] = \frac{r_0}{\lambda^2 D}\left[e^{-\lambda x} - \lambda x e^{-\lambda d} + 1 \right] \text{ (ans).}$$

(d) Oxygen diffusion governing equations :-

(a) $r^2 \partial_r^2 [c] + r \partial_r [c] - r^2 \left(\sqrt{\frac{K}{D_s}}\right)^2 [c] = 0.$

$\left[N = -D \frac{\partial [c]}{\partial r} \right.$.

Let $\alpha = \sqrt{\frac{K}{D_s}}$.

$$\underbrace{r^2 \partial_r^2 [c] + r \partial_r [c] - (\alpha^2 r^2)[c] = 0}$$

Modified Bessel equation of order 0.

$[c] = A I_0(\alpha r) + B K_0(\alpha r)$.

0 → infinity (always positive)

(b) Use boundary values :-

$R_0 = -2\pi R N \Big|_{r=R}$.

$= -2\pi R (-D) \left(\frac{\partial [c]}{\partial r} \right) \Big|_{r=R}$.

$= \frac{-2\pi R (-D) [c_0]}{I_0(\alpha R)} \cdot \left[\frac{\partial}{\partial r} I_0(\alpha r) \right]$.

$\Rightarrow \frac{2\pi D R [c_0] \alpha \cdot I_1(\alpha R)}{I_0(\alpha R)}$.

$$\boxed{R_0 = \frac{2\pi R [c_0] \sqrt{KD}}{I_0\left(\sqrt{\frac{K}{D}} R\right)} \cdot I_1\left(\sqrt{\frac{K}{D}} R\right)}$$

$K_0(\alpha r) = \int_0^\infty \exp\left(-\sqrt{\frac{K}{D}} r \cosh t\right) dt$

$\therefore \exp(f(x)) \Big|_{\text{finite}}^{0} \longrightarrow \exp(f(x)) \; \infty$.

$\therefore K_0(\alpha r)$ blows up @ 0.

[$\because f(x)$ is an increasing func^n].

$\therefore K_0 \longrightarrow \infty$ as $r \longrightarrow 0$,

$K_0 \longrightarrow 0$ as $r \longrightarrow \infty$ $(e^{-\infty} \to 0)$.

$\therefore B = 0$ (MUST BE)

$\therefore [c] = A I_0(\alpha r)$.

$$\therefore \boxed{[c] = [c_0] \frac{I_0(\alpha r)}{I_0(\alpha R)}} \text{ (ans)}$$

We must do asymptotic analysis Now.

(c) Series expansions.

$$I_0\left(\sqrt{\frac{K}{D}}\cdot R\right) = 1 + \frac{\left(\sqrt{\frac{K}{D}}\cdot R\right)^2}{4} \ldots$$

$$I_1\left(\sqrt{\frac{K}{D}}R\right) = \sqrt{\frac{K}{D}}\frac{R}{2}.$$

$$\therefore \frac{I_1}{I_0} = \frac{\sqrt{\frac{K}{D}}\,R/2}{1 + \sqrt{\frac{K}{D}}\,R^2/4}.$$

$$\therefore R_0 = \frac{2\pi[C_0]\sqrt{KD}\cdot R^2/2}{1 + \sqrt{\frac{K}{D}}\cdot R^2/4}.$$

$$\therefore R_0 = g(R).$$

$$\lim_{R\to 0} g(R) \Rightarrow \lim_{R\to 0} R_0(R) = 0.$$

$$\lim_{R\to\infty} g(R) = \lim_{R\to 0} \left(2\pi[C_0]\cancel{\sqrt{KD}}\right)\cdot\frac{\cancel{R^2/2}}{\cancel{\sqrt{\frac{K}{D}}}\cdot R^2/\cancel{4}_2}$$

$$R_0 = 4\pi D[C_0]$$

$\therefore$ At very large R, $R_0 = 4\pi D[C_0]$.

At very small R, $R_0 = 0$.

Asymptotic Analysis.

Now, Behavior near 0.

$$R_0 = \left[2\pi [C_0] \sqrt{KD} \left(\frac{R^2/2}{1 + \sqrt{\frac{K}{D}} R^2/4} \right) \right].$$

Let $\pi [C_0] \sqrt{KD}$ be γ and $\frac{1}{4}\sqrt{\frac{K}{D}}$ be θ.

$$\therefore R_0 = \frac{\gamma R^2}{1 + \theta R^2}$$

$$\therefore R_0 = \frac{\gamma}{\theta}\left[\frac{(\theta R^2 + 1) - 1}{\theta R^2 + 1}\right] = \frac{\gamma}{\theta}\left\{1 - \frac{1}{1+\theta R^2}\right\}$$

We know γ & θ are both $\mathbb{R}^+$.

$\therefore$ Plot should have 0 @ $R = 0$

Plot should have $\gamma/\theta = 4\pi D [C_0]$ @ $R \rightarrow \infty$.

$R_0(R)$ is continuous & differentiable & smooth.